

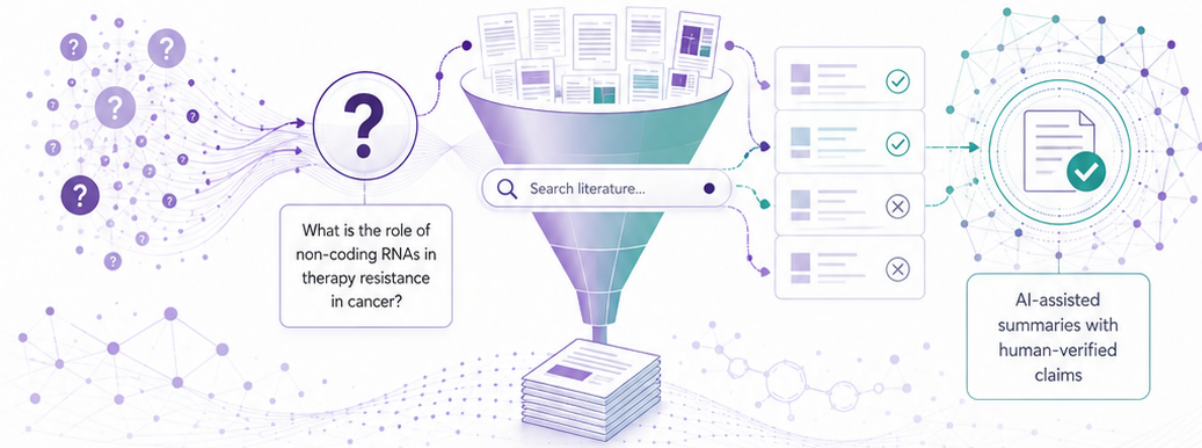


WEEK 2 READING MATERIAL

Scientific Question Formulation and Literature Mining with AI

From curiosity to search strategy, evidence synthesis, and human-verified claims

BROAD CURIOSITY → REFINED QUESTION → LITERATURE SEARCH → EVIDENCE SCREENING → VERIFIED INSIGHTS



QUESTION DESIGN

Transform curiosity into focused, answerable, and researchable questions.

non-coding RNA AND
therapy resistance AND cancer

MeSH Keywords Synonyms Filters



SEARCH STRATEGY

Build effective queries using controlled vocabulary, Boolean logic, and filters.



EVIDENCE SYNTHESIS

Extract, organize, and summarize key findings across studies.



HUMAN VERIFICATION

Critically appraise evidence and confirm claims with human expertise.



Fall 2026

Two 40-Minute Undergraduate Sessions

Dr. Liwei Xie

Week 2 learning architecture

From curiosity to a verified, evidence-based bioinformatics question



Three habits students practice this week

Ask before analyzing

Define the comparison and outcome before touching data.

Search before claiming

Use PubMed, MeSH, Boolean logic, and citation networks before judging novelty.

Verify before citing

Treat AI-generated literature summaries as drafts until checked against real sources.

Figure 1. Week 2 learning architecture: students move from curiosity to a verified, evidence-based, data-driven bioinformatics question.

How to use this handout

Read Session 1 before class 1 and Session 2 before class 2. During class, use the tables and prompt templates as working tools. After both classes, return to the exercises and reference list when preparing your own mini research question and literature-mining workflow.

Learning objectives and deliverables

By the end of Week 2, students should be able to refine a broad biological topic into a feasible bioinformatics question, construct a PubMed search strategy, use AI to expand and screen literature without surrendering judgment, and organize references in a reproducible library.

Competency	What students should be able to do	Concrete deliverable
Question formulation	Convert a broad topic into a specific, testable, data-driven question	A six-level question funnel
Research design	Use PICO/PECO and feasibility criteria to define population, exposure, comparison, outcome, data source, and validation	A one-page analysis plan
Literature mining	Build PubMed queries using MeSH terms, Boolean operators, field tags, and date filters	A documented search string with result count
AI-assisted synthesis	Use LLMs for query expansion, screening, summarization, and gap discovery while verifying citations	A prompt log and a source-verification table
Reference management	Organize papers using Zotero or another reference manager	A small library with 5 annotated papers

Class schedule

Session	Main focus	In-class activity	Homework connection
Session 1: 40 min	From vague interest to testable research question	Question funnel + PICO mini-workshop	Draft a data-driven question
Session 2: 40 min	AI-assisted literature mining and source verification	PubMed query clinic + AI prompt clinic	Find and annotate 3-5 papers

Session 1. The art and science of formulating research questions

Good bioinformatics begins before code is written. It begins when a student can say exactly what biological relationship is being tested, which samples or systems are relevant, which measurements are needed, and what result would count as evidence. Without this discipline, large public datasets can become traps: one can test many patterns, obtain impressive-looking figures, and still fail to produce a defensible scientific claim.

The purpose of Session 1 is to make question formulation concrete. Students will learn to move from a vague domain to a data-driven question, use PICO or PECO to structure the comparison, identify common design mistakes, and turn a question into a minimal analysis plan. These skills prepare students for GEO/NCBI data retrieval, RNA-seq analysis, microbiome analysis, and multi-omics integration later in the course.

Core principle

A good bioinformatics research question is specific, testable, biologically motivated, feasible with available data, and interpretable through a planned computational workflow.

1.1 Why vague questions produce weak science

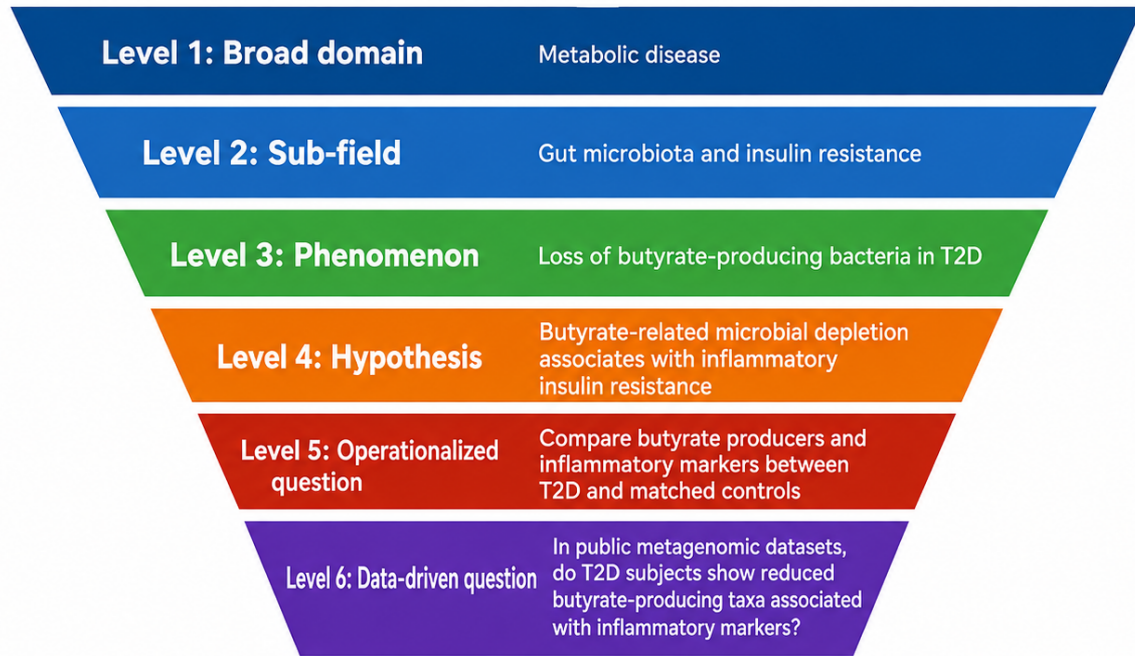
A vague question is not merely incomplete; it changes the behavior of the analyst. If the question is “How does the microbiome affect health?”, nearly any dataset, any outcome, and any plot can appear relevant. The analysis becomes exploratory without boundaries. Exploration is valuable, but it should not be disguised as confirmatory evidence. In the literature, this problem is often connected to selective analysis, multiple testing without correction, and “p-hacking,” where one searches for significance after seeing the data [1,2].

In bioinformatics, the problem is amplified by data scale. A single RNA-seq dataset may contain tens of thousands of genes, a metagenomic dataset may contain thousands of taxa or gene families, and a clinical cohort may include dozens of covariates. Every extra feature creates more chances to find a low p-value by accident. For this reason, question formulation is not a soft skill; it is a statistical safeguard and a scientific integrity practice.

1.2 The question formulation funnel

The Question Formulation Funnel is a practical model for narrowing a broad area into a bioinformatics question that can actually be analyzed. Each level adds specificity. The funnel should be used before selecting a dataset or writing code. Students should be able to explain why each narrowing step is biologically meaningful and how it changes the downstream analysis plan.

Scientific question formulation funnel



Classroom rule: stop narrowing only when the question implies a population, a comparison, a measurable outcome, and a feasible dataset.

Figure 2. The Scientific Question Formulation Funnel. Each level adds biological and analytical specificity until the question can be mapped to data, methods, and outputs.

Level	Question state	Student task	Example
1. Broad domain	General area of interest	Name the field or disease area	Cancer biology; metabolic disease; host-microbiome interaction
2. Sub-field	Narrower biological context	Specify tissue, disease, system, or pathway	Gut microbiota and type 2 diabetes
3. Phenomenon	Specific observation or pattern	Identify what has been observed but not fully explained	Reduced butyrate-producing bacteria in insulin resistance
4. Testable hypothesis	Falsifiable claim about relationship or mechanism	State direction, condition, and expected relationship	Loss of butyrate pathways associates with inflammatory transcriptional signatures
5. Operationalized question	Concrete measurement or comparison	Define groups, variables, and outcome	Are butyrate-producing taxa lower in T2D than controls, after adjusting for age and BMI?

6. Data-driven question	Question matched to available data and analysis	Name dataset, data type, method, and output	In public metagenomic datasets, do butyrate pathway abundances correlate with host inflammatory genes?
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1.3 From broad topic to data-driven question: worked examples

The examples below illustrate how a broad topic can be sharpened into a usable bioinformatics question. Notice that the final question includes a biological relationship, data type, comparison, and expected analytical output.

Broad topic	Focused biological issue	Data-driven question	Likely analysis
Gut microbiota and obesity	Microbial metabolites and insulin resistance	In adults with obesity, are microbial butyrate biosynthesis pathways associated with fasting insulin after adjusting for diet and BMI?	Metagenomic pathway profiling + regression
Cancer immunology	Single-cell tumor microenvironment	In single-cell RNA-seq data from colorectal cancer, are exhausted CD8+ T cell signatures enriched in tumors with high myeloid inflammation?	Cell annotation + module scoring + group comparison
Muscle regeneration	Satellite cell activation	After muscle injury, does a specific transcription factor show early activation in satellite cells before differentiation markers rise?	Time-course RNA-seq or scRNA-seq trajectory analysis
Antibiotic resistance	Resistance gene transfer	Are beta-lactam resistance genes enriched in hospital-associated gut metagenomes compared with community controls?	AMR gene profiling + differential abundance
MAFLD/MASH biomarkers	Secreted proteins and disease severity	Which plasma proteins distinguish MAFLD from MASH and remain associated after adjusting for BMI and age?	Differential protein abundance + ROC analysis

1.4 PICO, PECO, and FINER: three useful frames

PICO is widely used in biomedical research to structure clinical and translational questions [3]. In observational bioinformatics, PECO is often more natural: Population, Exposure, Comparison, Outcome. For example, “exposure” may be high expression of a gene, presence of a mutation, abundance of a microbial taxon, or a treatment condition. FINER is a complementary quality check: a good question should be feasible, interesting, novel, ethical, and relevant.

Framework	Elements	Best use in this course	Example
PICO	Population, Intervention/Exposure, Comparison, Outcome	Clinical or biomedical questions with explicit comparison	In CRC patients, does high LDHA expression compared with low expression associate with poor survival?
PECO	Population, Exposure, Comparison, Outcome	Observational omics questions	In obese adults, does low butyrate-pathway abundance

			compared with high abundance associate with insulin resistance?
FINER	Feasible, Interesting, Novel, Ethical, Relevant	Quality check before analysis	Can public data answer this? Is the question not already settled? Does it matter biologically?

PICO converts a research question into an analysis plan

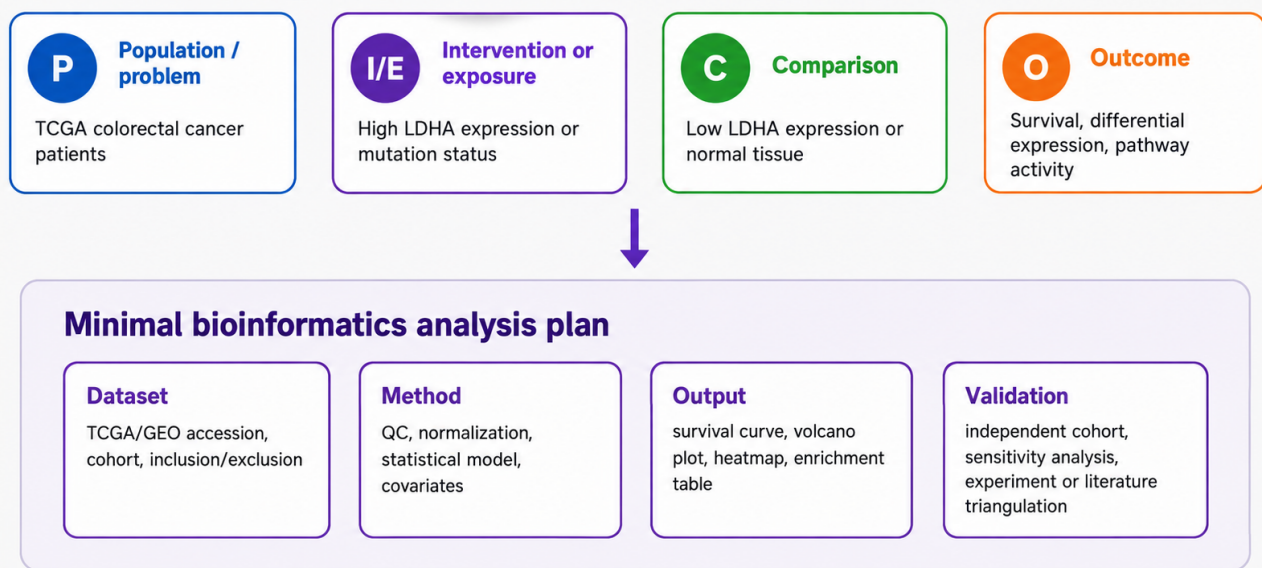


Figure 3. PICO/PECO helps students convert a biological question into a minimal analysis plan with data source, method, output, and validation.

1.5 The minimal analysis plan

Once the question is formulated, the next step is not coding. The next step is a minimal analysis plan. This plan is a short document that explains how the question will be answered and what assumptions the analysis depends on. It protects the project from drifting into uncontrolled exploration. It also makes collaboration easier because another student or instructor can inspect the plan and detect weak design before time is wasted.

Plan component	Question to answer before analysis	Example
Dataset	Which cohort, accession, or public database will be used?	TCGA COAD; GEO GSE39582; HMP; curatedMetagenomicData
Sample inclusion	Which samples are included or excluded?	Primary tumor only; remove samples without survival data; adult subjects only

Primary comparison	What exactly is compared?	High vs low LDHA expression; T2D vs control; treated vs untreated
Covariates	What confounders must be considered?	Age, sex, BMI, batch, sequencing depth, tissue site
Method	What statistical model or workflow is appropriate?	DESeq2 design formula; Cox regression; PERMANOVA; Wilcoxon test
Visualization	What figure will show the result clearly?	PCA, volcano plot, survival curve, heatmap, forest plot
Validation	What will increase confidence?	Independent cohort, sensitivity analysis, literature triangulation, perturbation experiment

1.6 Common pitfalls in question formulation

Pitfall	Why it is problematic	Weak version	Better version
Too broad	No single dataset or analysis can answer it	How does the microbiome affect health?	Which microbial pathways differ between T2D patients and matched controls?
Not testable	No measurable variable or comparison	Is inflammation important?	Are IL6 and TNF pathway scores higher in diseased tissue than controls?
Confounded	Observed signal may reflect other variables	Are obese and lean microbiomes different?	Are microbial differences still present after adjusting for age, sex, diet, and batch?
Reverse causality	Direction of relationship is unclear	Does cancer cause gene X expression?	Is high gene X expression associated with tumor stage and survival?
Trivially answered	No meaningful novelty remains	Does TP53 mutate in cancer?	Does TP53-mutant CRC show a distinct immune microenvironment signature?
Not feasible	Required data do not exist or are inaccessible	Analyze the proteome of every neuron in 10,000 brains	Use available single-nucleus RNA-seq datasets to profile selected cell types

Instructor note

When students present a research question, ask four questions: What is compared? What is measured? What dataset can answer it? What would count as a meaningful result?

1.7 In-class activity: question refinement workshop

Students should choose one of the broad topics below or propose their own. The goal is not to find a perfect research project in ten minutes; the goal is to practice making vague ideas more testable.

- Gut microbiota and metabolic disease
- Single-cell transcriptomics in cancer immunology
- Antibiotic resistance genomics
- Epigenetic regulation in neurological disorders
- Muscle regeneration and aging
- Proteomic biomarkers in MAFLD/MASH

Step	Student action	Output
1	Write a broad topic	One phrase
2	Narrow to a biological phenomenon	One sentence
3	State a testable hypothesis	One falsifiable claim
4	Apply PICO/PECO	Four structured elements
5	Identify a possible dataset or data type	One database, accession type, or measurement
6	Draft a minimal analysis plan	Data source, comparison, method, output, validation

Session 2. AI-assisted literature mining

Session 2 teaches students how to turn a research question into a structured literature-mining workflow. The central message is that AI can accelerate search, screening, summarization, and synthesis, but every AI-generated claim must be verified against real sources. Students should leave this class with a practical search strategy, prompt templates, a source-quality rubric, and a plan for building a small annotated reference library.

PubMed is still the core biomedical database for this course. It is a free resource from the U.S. National Library of Medicine that supports biomedical and life-sciences literature retrieval and includes more than 40 million citations and abstracts. PubMed does not provide full text for all papers, but links to PubMed Central, publisher pages, and other full-text sources are often available [4].

2.1 The literature landscape in bioinformatics

Biomedical literature grows faster than any student can read. Bioinformatics is particularly dense because it combines biology, medicine, statistics, databases, software, and computational methods. A single research topic may require knowledge of disease biology, sequencing technology, statistical models, public repositories, and prior validation studies. This makes literature mining a workflow rather than a one-time search.

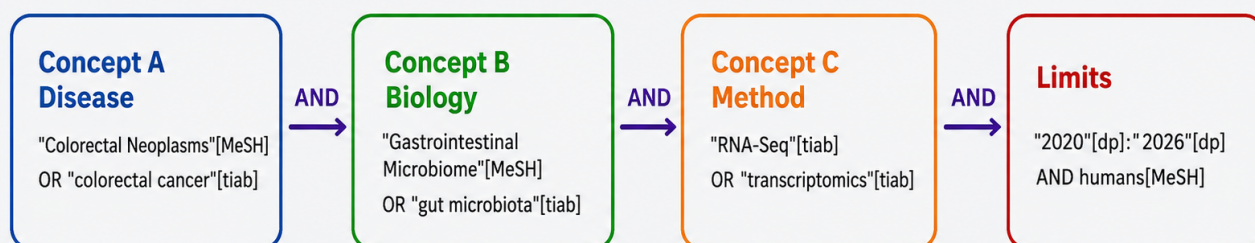
Students should distinguish four goals: finding papers, understanding papers, comparing papers, and discovering gaps. Traditional databases are strong for retrieval; AI systems can help summarize and structure information; citation-network tools help reveal intellectual neighborhoods; reference managers keep the process reproducible.

Tool	Strength	Best use	Caution
PubMed	Biomedical specificity, MeSH terms, advanced query syntax	Primary biomedical search	Does not cover all disciplines or always provide full text
Google Scholar	Broad coverage and citation counts	Finding highly cited or cross-disciplinary papers	Search syntax and ranking are less transparent
Semantic Scholar	AI-powered scientific search, TLDR, citation context	Finding related papers and quick summaries	AI summaries must still be checked
Elicit	AI-native extraction and literature-table workflows	Systematic question-driven screening	Not a replacement for full systematic review protocol
Perplexity / web AI search	Recent web-connected synthesis with source links	Fast orientation on emerging topics	Source quality varies; verify citations
Connected Papers / ResearchRabbit	Citation-network discovery	Mapping a field and expanding a reading list	Network proximity does not equal relevance
Zotero	Reference storage, PDF annotation, citation insertion	Course reference library	Requires consistent organization habits

2.2 PubMed search strategy: MeSH, Boolean logic, and field tags

A good PubMed search is built by concepts. First, identify the major concepts in your question. Second, list synonyms within each concept. Third, connect synonyms with OR. Fourth, connect different concepts with AND. Fifth, apply limits only after the core query is working. PubMed also supports MeSH terms, a controlled vocabulary that improves precision and recall because articles can be indexed under standardized biomedical terms [5].

Building a PubMed search string is a design problem



Final query structure

(Disease terms) AND (Microbiome terms) AND (RNA-seq/transcriptomic terms) AND (date/species limits)

Principle: use OR inside a concept to capture synonyms; use AND between concepts to narrow the search.

Figure 4. A PubMed query should be assembled concept by concept: OR expands synonyms within a concept, while AND narrows across different concepts.

Syntax	Meaning	Example	Teaching note
AND	Both concepts must appear	cancer AND microbiome	Narrows results
OR	Either synonym may appear	microbiome OR microbiota OR gut flora	Broadens within a concept
NOT	Exclude a concept	cancer NOT review	Use sparingly; can remove relevant papers
[tiab]	Search title and abstract	"gut microbiota"[tiab]	Useful for new terms not well indexed
[MeSH]	Search Medical Subject Heading	"Gastrointestinal Microbiome"[MeSH]	Useful for controlled biomedical concepts
[dp]	Date of publication	"2020"[dp]:'2026'[dp]	Useful for recent literature windows
[pt]	Publication type	review[pt] or clinical trial[pt]	Useful for filtering by evidence type

2.3 Worked example: gut microbiota, colorectal cancer, and transcriptomics

Suppose a student asks: “In colorectal cancer, are gut microbiome changes connected to host transcriptomic changes in the tumor or intestinal tissue?” A useful search should include disease terms, microbiome terms, and transcriptomics terms. A first PubMed query might be:

Example PubMed query

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("Colorectal Neoplasms"[MeSH] OR "colorectal cancer"[tiab] OR CRC[tiab]) AND
("Gastrointestinal Microbiome"[MeSH] OR "gut microbiota"[tiab] OR "gut microbiome"[tiab]) AND
("RNA-Seq"[tiab] OR transcriptomics[tiab] OR "gene expression profiling"[tiab]) AND
("2020"[dp]:"2026"[dp])
```

This query is not the final truth. It is a starting point. If it returns too few papers, remove one concept or broaden synonyms. If it returns too many papers, add tissue, method, species, or publication-type filters. Students should record each search string and result count so the workflow can be reproduced.

2.4 Using AI to accelerate literature mining

LLMs can support literature mining at multiple points: query expansion, translation of unfamiliar concepts, abstract screening, paper summarization, extraction of methods, and synthesis of gaps. The safest pattern is to use AI to generate candidate structure, then verify every key claim in real sources. AI should be treated as a fast but fallible assistant.

AI-assisted literature mining is an iterative, human-verified workflow

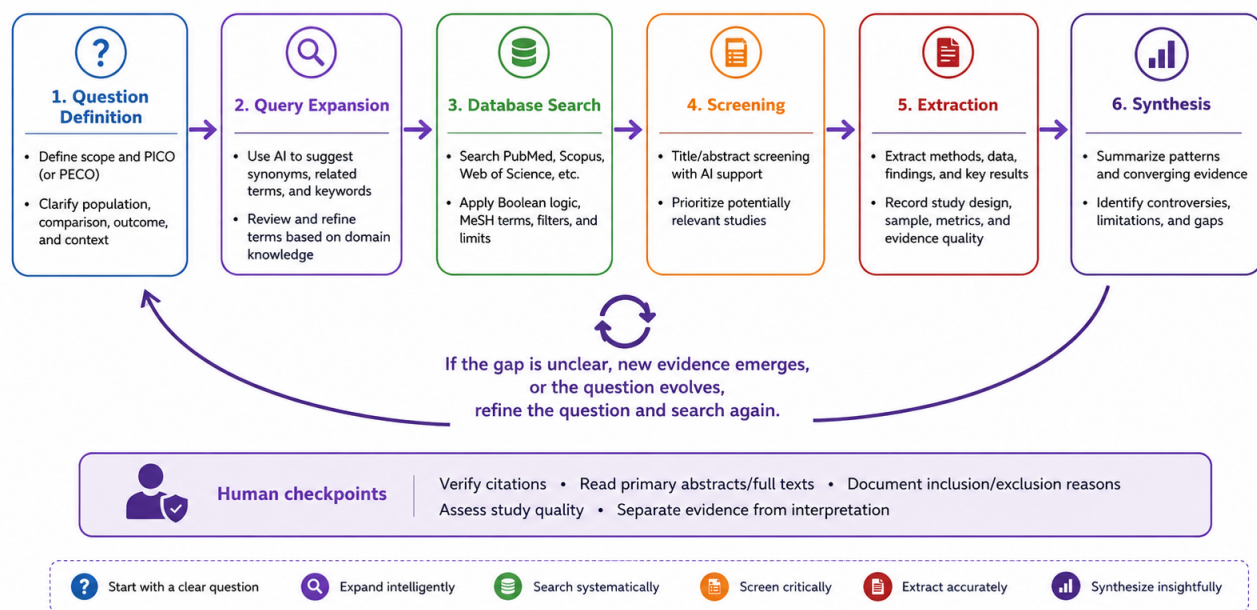


Figure 5. AI-assisted literature mining should be iterative and human-verified. AI is useful at several steps, but source checking and interpretation remain human responsibilities.

AI-supported task	Good prompt pattern	Human verification
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Query expansion	Generate synonyms, MeSH terms, and spelling variants for concept X	Check terms in PubMed MeSH browser and run test searches
Abstract screening	Classify abstracts as include/exclude/uncertain using explicit criteria	Manually review uncertain and high-impact papers
Paper summarization	Summarize research question, design, methods, findings, limitations	Read the abstract and figures; check whether summary matches the paper
Gap discovery	Compare summaries and identify consensus, controversy, and gaps	Confirm gaps by searching for recent reviews and primary studies
Hypothesis generation	Convert a literature gap into PICO/PECO and analysis plan	Assess feasibility and novelty before claiming importance

2.5 Citation hallucination and source verification

The most dangerous AI error in academic work is a fabricated citation. LLMs can generate plausible-sounding author names, titles, journal names, and DOI strings that do not exist. They can also cite a real paper for a claim that the paper does not support. For this reason, no citation should enter a class assignment, manuscript draft, or project proposal unless the student has verified that the source exists and has read at least the abstract.

Human-in-the-loop verification prevents AI-assisted literature errors

AI accelerates discovery, but human verification ensures accuracy and reliability.

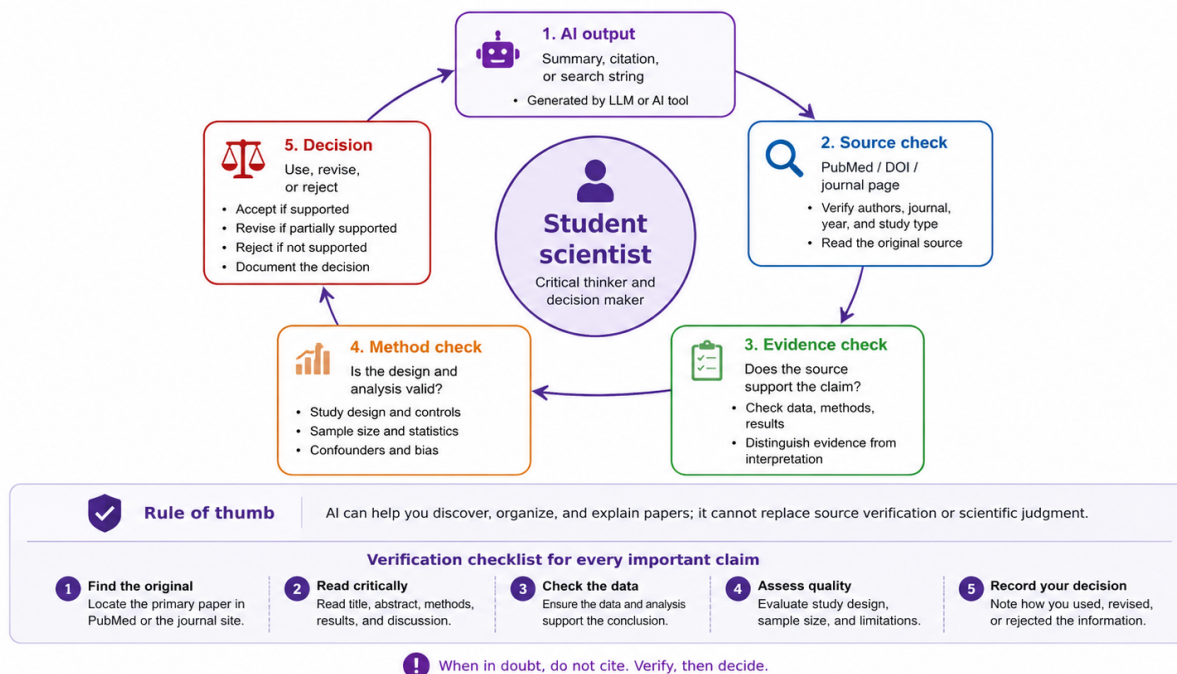


Figure 6. Human-in-the-loop verification helps students use AI without importing fabricated citations, unsupported claims, or misleading biological interpretations.

Risk	Example	How to detect	Required action
Fake citation	A plausible DOI that does not resolve	Search PubMed, Crossref, or journal site	Remove or replace with verified source
Misattributed claim	AI says a paper showed causality, but the study was observational	Read abstract, methods, and key figures	Downgrade claim to association or find validation
Wrong method description	AI says DESeq2 was used when paper used limma	Check methods section	Correct summary and note exact method
Outdated consensus	AI repeats older interpretation despite recent contradictory studies	Search recent reviews and 2024-2026 papers	Update synthesis and note controversy
Overconfident gap	AI claims nobody has studied a topic	Search exact gap terms and citation networks	Use cautious wording: "limited evidence" or "not yet well resolved"

2.6 Prompt templates for AI-assisted literature mining

The prompts below are designed for safe and useful AI-assisted literature work. Students should copy, adapt, and record them in a prompt log. The key is to provide context, define the task, specify the expected output format, and ask for uncertainty or verification steps.

Purpose	Reusable prompt
Query expansion	I am searching PubMed for papers about [topic]. Generate synonyms, related terms, MeSH terms, and one Boolean search string. Separate terms by concept and explain why each term is included.
Abstract screening	Use these inclusion criteria: [criteria]. Classify each abstract as INCLUDE, EXCLUDE, or UNCERTAIN. Provide one sentence of reasoning and do not invent information not present in the abstract.
Paper summary	Summarize this abstract using: research question, study design, data type, key methods, main finding, limitations, and relevance to my question.
Gap identification	Based on these paper summaries, identify consensus, controversy, and specific research gaps. For each gap, explain what evidence is missing.
Hypothesis design	Convert this literature gap into a PECO question, a testable hypothesis, a possible public dataset, and a minimal analysis plan.
Verification prompt	List every factual claim in your answer that requires source verification. For each claim, tell me what source type should verify it.

2.7 Source quality: not every paper should influence your conclusion equally

Source evaluation is a scientific skill. A high-quality bioinformatics paper usually has clear sample selection, adequate sample size, documented preprocessing, public data or accession numbers, code or software version information, appropriate multiple-testing correction, and validation. A weak paper may still be useful for hypothesis generation, but students should not treat all sources as equal evidence.

Criterion	Stronger evidence	Weaker evidence
Study design	Clear cohort definition, controls, covariates, and validation	Vague sample description or no controls

Sample size	Reasonable power for the analysis type	Very small n without justification
Data availability	Raw or processed data deposited in GEO/SRA/ENA/TCGA or equivalent	No accession number or unclear availability
Code/method transparency	Code, software versions, parameters, and pipelines documented	Only high-level method names provided
Statistics	Multiple-testing correction, effect sizes, covariate-aware models	Only unadjusted p-values or unclear tests
Biological interpretation	Claims match evidence and acknowledge limitations	Overstated mechanism from correlation
Reproducibility	Independent cohort, sensitivity analysis, or experimental validation	Single dataset with no validation

2.8 Building a reference library with Zotero

Reference management is part of reproducibility. Zotero is recommended because it is free, open-source, supports browser-based capture, stores PDFs, enables notes and tags, and integrates with Word or Google Docs for citation insertion [6]. Students should build their library as they search, not after writing begins.

Step	Action	Recommended habit
1	Install Zotero and browser connector	Create a course collection: Bioinformatics Week 2
2	Save papers directly from PubMed or journal pages	Check metadata after saving
3	Create subcollections	Question formulation, PubMed search, AI literature mining, topic-specific papers
4	Annotate PDFs	Highlight research question, dataset, method, finding, limitation
5	Add tags	method: RNA-seq; disease: CRC; evidence: review; data: TCGA
6	Export or cite	Use Word plugin or bibliography export for assignments

Post-class practical exercise

Complete this exercise after both Week 2 sessions. The task is designed to connect question formulation, literature mining, AI assistance, and source verification into one coherent workflow.

Step	Task	Required output
1	Choose a broad topic from the list or your own research interest	One broad topic
2	Use the Question Formulation Funnel	Six levels ending in a data-driven question
3	Apply PICO or PECO	Population, exposure/intervention, comparison, outcome
4	Write one PubMed query using MeSH, Boolean operators, and field tags	Search string + result count + date searched
5	Use an LLM for query expansion	Prompt used + revised search string
6	Screen and select 3-5 papers	Inclusion reason for each selected paper
7	Save papers in Zotero or equivalent library	Collection name + screenshot or exported bibliography
8	Write a short synthesis	Consensus, controversy, gap, and one refined hypothesis
Criterion	Excellent	Needs improvement
Question specificity	Question includes population, comparison, measurement, outcome, and feasible data source	Question remains broad, vague, or not measurable
Search strategy	Uses synonyms, MeSH, Boolean logic, field tags, and records result count	Uses only simple keywords or does not document search
AI use	Prompt log is included; AI output is checked and revised	AI output is copied without verification
Source quality	Selected papers are relevant, verified, and briefly evaluated	Papers are poorly matched or unverifiable
Synthesis	Identifies consensus, controversy, gap, and hypothesis	Only lists papers without synthesis

Self-assessment questions

1. What is the difference between a testable hypothesis and a data-driven question? Why does the distinction matter for bioinformatics?
2. Apply PICO or PECO to: “Does gut microbiota composition differ between type 2 diabetes patients and healthy controls, and can specific bacterial genera predict disease severity?”
3. Write a PubMed Boolean search string for RNA-seq analysis of the tumor microenvironment in breast cancer, published from 2020 to 2026.
4. What is citation hallucination? Describe how a fabricated AI citation could damage a student assignment or manuscript.
5. You find 200 abstracts after a PubMed search. How would you use an LLM to assist screening while maintaining scientific rigor?
6. Compare PubMed, Semantic Scholar, and Elicit. When would you use each tool?

Pre-reading for Week 3

Before the next class on public repositories and data cleaning, students should complete the following preparation tasks:

- Create or confirm an NCBI account.
- Browse GEO and record one dataset accession number in the format GSE#####.
- Open the dataset page and identify organism, platform, sample number, and data type.
- Confirm that R and RStudio are installed and working.
- Bring one biological question and one candidate public dataset to class.

Suggested reading

Category	Recommended reading	Why it matters
Research question design	Hulley et al., Designing Clinical Research; Richardson et al., PICO framework [3]	Useful for learning how questions become study designs
Statistical integrity	Ioannidis 2005; Simmons et al. 2011; Benjamini and Hochberg 1995 [1,2,7]	Explains false discovery, p-hacking, and multiple testing
Literature search	PubMed User Guide; NLM MeSH tutorial [4,5]	Required for reproducible biomedical searches
Systematic review thinking	PRISMA 2020 statement [8]	Introduces transparent search and screening workflows
AI and literature mining	Jin et al. 2024; recent reviews on AI-assisted biomedical literature search [9]	Frames PubMed and beyond in the AI era
Reference management	Zotero documentation [6]	Practical tool for project reference libraries

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One-page checklist for students

Before analysis	Yes/No
I can state my research question in one sentence.	
My question includes population/system, comparison/exposure, measurable outcome, and feasible data type.	
I have written a minimal analysis plan before downloading or analyzing data.	
I have built at least one PubMed query with synonyms, Boolean operators, and field tags.	
I recorded the date searched and number of results.	
I used AI for support but kept a prompt log.	
I verified all citations in PubMed, DOI lookup, or journal pages.	
I saved selected papers in Zotero or another reference manager.	
I can explain the consensus, controversy, and gap in my mini-topic.	

Appendix A. Worked case study: LDHA expression in colorectal cancer

This worked case shows how a student can move from a textbook-level cancer metabolism idea to a concrete bioinformatics question. It is intentionally simple enough for undergraduates, but it preserves the logic used in real research planning: define the biological mechanism, state the measurable exposure, select a comparison, identify outcomes, and decide what evidence would strengthen or weaken the claim.

Broad topic: cancer metabolism. Focused phenomenon: enhanced glycolysis in colorectal cancer. Candidate molecular factor: LDHA, a glycolytic enzyme frequently discussed in relation to lactate production and tumor metabolism. The student should not immediately download data. Instead, the student first asks whether the question is novel, whether public data exist, whether the outcome is measurable, and whether the analysis can distinguish association from mechanism.

Funnel level	Student output
Broad domain	Cancer metabolism
Sub-field	Colorectal cancer glycolysis
Phenomenon	High lactate metabolism may mark aggressive tumors
Hypothesis	High LDHA expression is associated with poorer survival and an inflammatory or hypoxic transcriptomic signature in colorectal cancer
Operationalized question	Are tumors with high LDHA expression different from tumors with low LDHA expression in survival and pathway activity?
Data-driven question	In TCGA COAD/READ RNA-seq and clinical data, does high LDHA expression associate with overall survival and glycolysis/hypoxia pathway enrichment after considering stage?
PICO/PECO element	Case-study definition
Population	Patients with colorectal adenocarcinoma in TCGA COAD/READ
Exposure	High LDHA expression, defined by upper quartile or median split
Comparison	Low LDHA expression group
Outcome	Overall survival, pathway scores, differential expression profile
Covariates	Tumor stage, age, sex, batch or platform when available
Interpretation boundary A TCGA analysis can show association, prognostic relevance, and pathway co-variation. It cannot prove that LDHA causes tumor progression without perturbation experiments such as knockdown, overexpression, isotope tracing, or animal validation.	

A suitable literature-mining step would search for colorectal cancer, LDHA, glycolysis, survival, and RNA-seq. A suitable analysis plan would include expression preprocessing, group definition, survival analysis, differential expression, pathway enrichment, and independent validation or literature

triangulation. A strong student report should explicitly say what is supported, what is speculative, and what experiment would be needed next.

Appendix B. Worked case study: gut microbiota, type 2 diabetes, and insulin resistance

This case study demonstrates how to work with a host-microbiome question. Microbiome topics are attractive to students because they are biologically rich, but they are also easy to overinterpret. Diet, medication, age, BMI, geography, sequencing method, and batch can all influence microbiome profiles. A strong question must therefore include confounders and avoid claiming causality from cross-sectional associations.

Design element	Recommended formulation
Broad topic	Gut microbiota and metabolic disease
Focused issue	Microbial pathways associated with insulin resistance in adults
Population	Adults with type 2 diabetes or obesity-related insulin resistance
Exposure	Reduced abundance of butyrate-producing taxa or butyrate biosynthesis pathways
Comparison	Matched controls or low-risk subjects
Outcome	Fasting insulin, HOMA-IR, inflammatory markers, or host transcriptomic pathway scores
Data type	16S, metagenomics, metabolomics, or paired host transcriptomics when available
Analysis	Alpha/beta diversity, differential abundance, pathway profiling, covariate-aware regression

A student might begin with the question: “Are butyrate-producing microbial taxa reduced in type 2 diabetes?” A stronger version would be: “In adult metagenomic datasets with BMI and medication metadata, are butyrate biosynthesis pathways lower in type 2 diabetes than matched controls, and do they associate with inflammatory host signatures?” The stronger question is longer, but it is more scientific because it names the population, data type, confounders, comparison, and biological interpretation.

Literature-search concept	Example terms
Disease	"Diabetes Mellitus, Type 2"[MeSH], type 2 diabetes[tiab], T2D[tiab]
Microbiome	"Gastrointestinal Microbiome"[MeSH], gut microbiota[tiab], intestinal microbiome[tiab]
Function	butyrate[tiab], short-chain fatty acid[tiab], SCFA[tiab], microbial pathway[tiab]
Data type	metagenomics[tiab], shotgun sequencing[tiab], 16S[tiab], metabolomics[tiab]
Common mistake Students often write “the microbiome causes diabetes.” A safer evidence-based formulation is “specific microbiome features are associated with diabetes-related phenotypes and may motivate mechanistic validation.”	

Appendix C. Worked case study: single-cell transcriptomics in the tumor microenvironment

Single-cell RNA-seq is powerful because it separates mixed tissues into cell states and cell types. However, single-cell questions require careful formulation. Students should ask whether the question concerns cell composition, gene expression within a cell type, cell-state transitions, ligand-receptor communication, spatial localization, or response to treatment. Each sub-question implies a different analysis.

Question type	Example question	Likely output
Cell composition	Are exhausted CD8 ⁺ T cells more abundant in high-inflammatory tumors?	Cell proportion comparison and stacked bar plot
Within-cell expression	Is PDCD1 higher in CD8 ⁺ T cells from non-responders?	Violin plot or pseudobulk differential expression
Cell-state program	Do T cells show enrichment of exhaustion or interferon-response signatures?	Module score or gene set score plot
Cell-cell communication	Do myeloid cells show increased ligand signaling toward T cells?	Ligand-receptor network or heatmap
Spatial context	Are immune-suppressive cells closer to tumor regions?	Spatial neighborhood analysis

A data-driven question might be: “In public single-cell RNA-seq datasets of colorectal cancer, are exhausted CD8⁺ T cell signatures enriched in tumors with high myeloid inflammatory signatures compared with low-myeloid tumors?” This question does not merely ask whether exhaustion exists; it defines a measurable immune context and a comparison.

Planning step	Student decision
Dataset	Search GEO, Single Cell Portal, or literature for human CRC scRNA-seq datasets
Cell annotation	Use published labels when available; otherwise annotate with canonical markers
Analysis unit	Prefer pseudobulk or sample-level summaries for group comparisons when possible
Quality control	Remove low-quality cells, doublets, and unassigned cells using documented criteria
Validation	Compare with independent dataset or published marker signatures
Interpretation boundary Single-cell clustering can identify cell states and associations, but it cannot by itself prove a causal interaction between cell types. Claims about mechanism require perturbation, spatial confirmation, or functional validation.	

Appendix D. Extended AI prompt library for Week 2

Students may use the following prompts as templates. Every prompt should be adapted to the actual topic, dataset, and assignment. Students should record the exact prompt used, the AI output, what they accepted, what they rejected, and how they verified key claims.

Task	Prompt template	Expected output
Question refinement	I am interested in [broad topic]. Help me narrow it through six levels: broad domain, sub-field, phenomenon, testable hypothesis, operationalized question, and data-driven question. Ask me clarification questions if needed.	Six-level funnel
PICO/PECO design	Convert this question into PICO or PECO. Identify missing elements and suggest how to make it more testable: [question].	Structured design table
Feasibility check	Evaluate whether this bioinformatics question is feasible using public data. List possible databases, data types, limitations, and what validation would still be needed.	Feasibility report
PubMed query expansion	Generate synonyms, MeSH terms, and a PubMed Boolean query for [topic]. Separate terms by concept and explain each field tag.	Search strategy
Search refinement	My PubMed query returned [number] results. Suggest how to broaden or narrow it without losing the main concept.	Revised search strings
Abstract screen	Using these inclusion criteria [criteria], classify each abstract as include, exclude, or uncertain. Give one sentence reason and do not use outside knowledge.	Screening table
Paper summary	Summarize this abstract in six fields: research question, design, data, methods, findings, limitations.	Structured summary
Gap synthesis	Here are summaries of selected papers. Identify consensus, controversy, and gaps. Do not invent papers or unsupported claims.	Gap table
Citation verification	List which claims in your answer require citation verification and suggest how I should verify each one.	Verification checklist
Hypothesis generation	Based on this gap, generate three testable hypotheses and rank them by feasibility for an undergraduate bioinformatics project.	Ranked hypotheses

Appendix E. Student worksheets

These worksheets can be copied into a course notebook, filled during class, or submitted as part of the Week 2 assignment. The goal is not to produce a perfect project proposal; the goal is to document the reasoning path from topic to question to evidence.

Question funnel worksheet	Student response
Broad domain	
Sub-field	
Specific phenomenon	
Testable hypothesis	
Operationalized question	
Data-driven question	
Possible data source	
Expected output figure	
Validation or limitation	
PICO/PECO worksheet	Student response
Population/problem	
Intervention or exposure	
Comparison	
Outcome	
Key covariates	
Data type	
Analysis method	

Appendix F. Search and source-extraction templates

Search log field		Student entry	
Database searched			
Date searched			
Search string			
Filters used			
Number of results			
Reason for refining search			
Final query selected			
Paper extraction field	Paper 1	Paper 2	Paper 3
Citation verified?			
Study question			
Data type and sample size			
Key method			
Main finding			
Limitation			
Relevance to my question			
Quality concern			
AI verification log		Student entry	
Prompt used			
AI output accepted			
AI output rejected			
Citation checked in PubMed/DOI?			
Claim that required verification			
How the claim was verified			
Remaining uncertainty			

Appendix G. Glossary of Week 2 concepts

Term	Meaning in this course
Research question	A focused question that specifies what relationship, comparison, or mechanism is being investigated
Hypothesis	A falsifiable statement that predicts a relationship or mechanism
Data-driven question	A research question translated into available data, variables, and analysis outputs
PICO	Population, Intervention/Exposure, Comparison, Outcome
PECO	Population, Exposure, Comparison, Outcome; often useful for observational omics studies
FINER	Feasible, Interesting, Novel, Ethical, Relevant
MeSH	Medical Subject Headings, the controlled vocabulary used by PubMed
Boolean operator	AND, OR, NOT operators used to combine search concepts
Field tag	A PubMed syntax marker such as [tiab], [MeSH], [dp], or [pt]
Data dredging	Uncontrolled searching for patterns after seeing data
p-hacking	Analytical flexibility that can produce false-positive significance
Multiple testing	Performing many statistical tests, increasing false discovery risk
FDR	False discovery rate, often controlled by Benjamini-Hochberg correction
Citation hallucination	AI-generated citation that is fake or does not support the claim
Source triangulation	Checking a claim across primary papers, reviews, databases, and methods documentation
Reference manager	Software such as Zotero that organizes papers, PDFs, notes, and citations
Screening criteria	Rules used to include, exclude, or mark literature as uncertain
Evidence synthesis	Combining findings across papers to identify consensus, controversy, and gaps
Human-in-the-loop	A workflow in which AI assists but human users verify and decide
Analysis plan	Pre-specified data, methods, outputs, and validation steps used to answer a question

Appendix H. Mini-review writing template

The Week 2 workflow is not complete until the student can convert search results into a short evidence-based mini-review. This template can be used for the final paragraph of the Week 2 assignment or as the opening logic for a semester project. The writing should be concise, evidence-aware, and careful about uncertainty.

Mini-review section	Guiding question	Suggested length
Background	What biological problem motivates the topic?	3-4 sentences
Current evidence	What do the selected papers consistently show?	4-6 sentences
Conflict or uncertainty	Where do papers disagree or remain incomplete?	2-4 sentences
Research gap	What specific issue remains unresolved and why does it matter?	2-3 sentences
Proposed question	What data-driven question follows from the gap?	1 sentence
Proposed analysis	Which dataset, method, figure, and validation would address it?	3-5 sentences
Writing rule Do not write “the literature proves.” Instead, write with calibrated language: “current evidence suggests,” “several studies report,” “the mechanism remains unresolved,” or “this association requires validation.”		

Example closing logic: “Together, these studies suggest that microbiome-derived short-chain fatty acid pathways may be altered in metabolic disease. However, most studies are cross-sectional and differ in sequencing platform, diet metadata, and medication control. A feasible bioinformatics question is therefore whether butyrate biosynthesis pathways remain associated with insulin-resistance phenotypes after covariate adjustment in public metagenomic cohorts. This question can be addressed by pathway profiling, regression modeling, and validation in an independent cohort.”

Appendix I. Responsible AI use policy for Week 2 assignments

AI use is encouraged in this course because modern bioinformatics increasingly depends on AI-assisted search, coding, debugging, and synthesis. However, students must disclose AI assistance and verify all scientific claims. The purpose of this policy is not to restrict learning; it is to make AI use transparent, reproducible, and scientifically accountable.

Allowed AI use	Required student responsibility
Generate synonyms, MeSH terms, or search strings	Run the search in PubMed and record the actual result count
Summarize abstracts or papers	Read the abstract and check whether the summary is accurate
Screen abstracts using criteria	Manually review uncertain or high-impact papers
Suggest gaps or hypotheses	Verify novelty by additional searches and recent reviews
Draft R or Python code for later assignments	Run code, inspect output, and explain each line
Improve wording	Ensure the content reflects verified evidence and your own interpretation
Not acceptable	Why it is not acceptable
Citing a paper suggested by AI without verifying it	Citation hallucination can create nonexistent references
Submitting AI-generated synthesis without reading sources	The student cannot judge whether the synthesis is accurate
Allowing AI to decide inclusion/exclusion without criteria	Screening becomes opaque and non-reproducible
Using AI to invent methods, data, or results	This is academic misconduct and damages scientific trust
Required disclosure sentence “AI tools were used for [query expansion / abstract screening / language polishing / code drafting]. All citations were verified in PubMed, DOI lookup, or journal pages, and all final interpretations were checked against the original sources.”	

Appendix J. Example final Week 2 submission outline

A strong Week 2 submission should be short, structured, and reproducible. The outline below can be used as a template. Students should not submit long uncontrolled essays. The value of the assignment comes from transparent reasoning and verifiable evidence.

Section	Expected content	Length
1. Broad topic and refined question	Six-level funnel ending in one data-driven question	Half page
2. PICO/PECO table	Population, exposure/intervention, comparison, outcome, covariates	One table
3. PubMed search strategy	Original query, AI-expanded query, result counts, date searched	Half page
4. Selected papers	3-5 verified papers with one-sentence relevance notes	One table
5. Evidence synthesis	Consensus, controversy, and gap	Half page
6. Proposed analysis plan	Dataset, comparison, method, output figure, validation	Half page
7. AI disclosure	Prompt log and verification statement	Short paragraph
Quality target A good submission should allow another student to reproduce the search, understand why papers were selected, identify what AI helped with, and see how the final research question follows from verified literature rather than from guesswork.		